

MCQs:-

1. B

2. C

3. B

4. C

5. B

6. A

7. C

8. B

9. A

10. C

11. B

12. B

$$Q: 3(1)(01)$$

$$A: 3(1)(01)$$

$$\log n = \frac{\sqrt[3]{8.59} \times (55.6)^2}{2.51 \times \sqrt{2.12}}$$

$$= \frac{(8.59)^{\frac{1}{3}} \times (55.6)^2}{2.51 \times \sqrt{2.12}}$$

$$\log x = \frac{(8.59)^{\frac{1}{3}} \times (55.6)^2}{2.51 \times (2.12)^{\frac{1}{2}}}$$

$$= \log 8.59^{\frac{1}{3}} + \log 55.6^2 - \log 2.51 - \log 2.12^{\frac{1}{2}}$$

$$\frac{1}{3} \log 8.59 + 2 \log 55.6 - \log 2.51 - \frac{1}{2} \log 2.12$$

$$\frac{1}{3}(0.9339) + 2(1.7450) - 0.3996 - \frac{1}{2}(0.3263)$$

$$0.3113 + 3.49 - 0.3996 - 0.16315$$

$$3.23855$$

$$\text{Antilog } 3.23855 \approx 1732$$

$$Q: 3(ii)(or)$$

$$A: 3(ii)(or)$$

$$(a) 4a^2 + 12ab + 9b^2$$

$$(2a)^2 + 2(2a)(3b) + (3b)^2$$

$$= (2a + 3b)^2$$

$$(b) (x^2 + 3x - 4)(x^2 + 3x + 5) + 8$$

$$\text{let } x^2 + 3x = a$$

$$(a - 4)(a + 5) + 8$$

$$= a(a + 5) - 4(a + 5) + 8$$

$$a^2 + 5a - 4a - 20 + 8$$

$$a^2 + a - 12$$

$$a^2 - 3a + 4a - 12$$

$$a(a - 3) + 4(a - 3)$$

$$(a + 4)(a - 3)$$

$$(x^2 + 3x + 4)(x^2 + 3x - 3)$$

$$(c) 8a^3 - 125b^3 - 2a + 5b$$

$$(2a)^3 - (5b)^3 - 2a + 5b$$

$$(2a - 5b)((2a)^2 + (2a)(5b) + (5b)^2) - 2a + 5b$$

$$(2a - 5b)(4a^2 + 10ab + 25b^2) - (2a - 5b)$$

$$(2a - 5b)(4a^2 + 10ab + 25b^2 - 1)$$

Q: 3 (ii)
A: 3 (iii)

$$\frac{y+1}{3} + \frac{y+1}{2} = 2 - \frac{y+3}{2}$$

$$\frac{y+1}{3 \times 2} + \frac{y+1}{2 \times 3} = \frac{2 \times 2 - y + 3}{1 \times 2 \times 2}$$

$$\frac{2y+2+3y+3}{6} = \frac{4-(y+3)}{2}$$

$$\frac{2y+3y+3+2}{6} = \frac{4-y-3}{2}$$

$$\frac{5y+5}{6} = \frac{4-3-y}{2}$$

$$\frac{5y+5}{6} = \frac{1-y}{2}$$

$$6(1-y) = 2(5y+5)$$

$$6 - 6y = 10y + 10$$

$$6 - 10 = 10y + 6y$$

$$-4 = 16y$$

$$-4 \div 16 = y$$

$$-\frac{1}{4} = y$$

$$\textcircled{1} \left(\frac{3}{4}\right)x - 1 \geq x + 1$$

$$\frac{3}{4}x - 1 \geq x + 1$$

$$\frac{3x}{4} - x \geq +1 + 1$$

$$\frac{3x}{4} - \frac{x \times 4}{1 \times 4} \geq 2$$

$$\frac{3x - 4x}{4} \geq 2$$

$$\frac{-x}{4} \geq 2$$

$$-x \geq 2 \times 4$$

$$-x \geq 8$$

$$x \leq -8$$

$$\{x/x \in \mathbb{R}, x \leq -8\}$$

Q.5 (iv)

A: 3 (iv)

A(0,0) B(6,0) C(6,4) D(0,4)

$d_{AB} =$

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(6-0)^2 + (0-0)^2}$$

$$= \sqrt{(6)^2 + (0)^2}$$

$$= \sqrt{36 + 0}$$

$$= \sqrt{36}$$

$$= 6$$

$$d_{BC} = \sqrt{(6-6)^2 + (4-0)^2}$$

$$= \sqrt{(0)^2 + (4)^2}$$

$$= \sqrt{0 + 16}$$

$$= \sqrt{16}$$

$$d_{CD} = \sqrt{(0-6)^2 + (4-4)^2}$$

$$= \sqrt{(-6)^2 + (0)^2}$$

$$= \sqrt{36 + 0}$$

$$= \sqrt{36}$$

$$= 6$$

$$d_{BD} = \sqrt{(0-6)^2 + (4-0)^2}$$

$$= \sqrt{(0)^2 + (4)^2}$$

$$= \sqrt{0 + 16}$$

$$= \sqrt{16}$$

$$4$$

$$AB = CD = 6$$

$$BC = AD = 4$$

These are not equal opposite side.

Diagonals:-

$$d_{AC} = \sqrt{(6-0)^2 + (4-0)^2}$$
~~$$= \sqrt{6^2 + 4^2}$$~~

$$= \sqrt{(6)^2 + (4)^2}$$

$$= \sqrt{36 + 16}$$
~~$$= \sqrt{100}$$~~

$$= \sqrt{52}$$

$$d_{BD} = \sqrt{(0-6)^2 + (4-0)^2}$$

$$= \sqrt{(-6)^2 + (4)^2}$$

$$= \sqrt{36 + 16}$$

$$= \sqrt{52}$$

$AC = BD$
equal ~~sides~~ diagonals.

Midpoint of AC :-

~~$$\frac{0+6}{2}, \frac{0+4}{2}$$~~

$$\frac{6}{2}, \frac{4}{2}$$

$$3, 2$$

midpoint of BD :-

$$A : \frac{6+0}{2}, \frac{4+0}{2}$$

$$\frac{6}{2}, \frac{4}{2}$$

$$3, 2$$

Q. 2 (i) (or)

A : 2 (i) (or)

(i) $(-8)^{\frac{1}{3}}$

$= ((-2)^3)^{\frac{1}{3}}$

$= -2 \text{ or } \frac{1}{2}$

(ii) $(\sqrt{25})^4$

~~$(5^{\frac{1}{2}})^4$~~ $(5)^4$

~~$(5)^2$~~ 625

iii) $(x^3)^2 \cdot (x^5)^{-1}$

$$x^{-6} \cdot 1$$

$$x^{-6} \quad \text{or} \quad \frac{1}{x^6}$$

_____ Q: 2(ii) _____
 _____ A: 2(ii) _____

a) True

b) True

c) False

example:- $\frac{1}{2}$ is a rational
 but not an integer.

_____ Q: 2(iii) (or) _____
 _____ A: 2(iii) (or) _____

i) 7.05×10^{-4}
 0.000705

ii) 2.31×10^6
 2310000

iii) 0.0000000512
 5.12×10^{-8}

Q: 2 (iv)

A: 2 (iv)

$$\textcircled{a} \log_x 125 = 3$$

$$125 = x^3$$

$$(5)^3 = x^3$$

$$5 = x$$

$$x = 5$$

$$\textcircled{b} \log_4 x = -\frac{3}{2}$$

$$x = (4)^{-\frac{3}{2}}$$

$$x = \left((2)^2\right)^{-\frac{3}{2}}$$

$$x = 2^{-3}$$

$$x = -8$$

$$\text{or } x = \frac{1}{8}$$

$$\textcircled{c} \log_2 (x^2 - 3) = \log_2 13$$

$$\log_2 (x^2 - 3) = \log_2 13$$

$$x^2 - 3 = 13$$

$$x^2 = 13 + 3$$

$$x^2 = 16$$

$$x^2 = (4)^2$$

$$x = 4$$

$$\text{Q: } 2(v) \text{ ---}$$

$$\text{A: } 2(v) \text{ ---}$$

i) 4580
 4.580×10^3
 characteristic :- 3
 mantissa :- $6608 + 1$
 6609
 3.6609

ii) 0.00093
 9.3×10^{-4}
 Characteristic :- $\bar{4}$
 mantissa :- $9684 + 1$
 9685
 $= \bar{4}.9685$

iii) 54
 5.4×10^1
 Characteristic :- 1
 mantissa :- $7323 + 1$
 7324
 1.7324

$$\text{Q: } 2(v_i) \text{ ---}$$

$$\text{A: } 2(v_i) \text{ ---}$$

i) $x^4 + 4m^4$
 $(x^2)^2 + (2m^2)^2$
 $(x^2)^2 + 2(x^2)(2m^2) + (2m^2)^2$
 $(x^2 + 2m^2)^2 - (2mx)^2$

$$(n^2 + 2m^2 + 2nm)(n^2 + 2m^2 - 2nm)$$

(b) $3x^2 + 11x + 6$
 $3x^2 + 9x + 2x + 6$
 $3x(x+3) + 2(x+3)$
 $(3x+2)(x+3)$

Q 2(vii)
 A 2(vii)

(a) $x^2 - 49$
 $(x)^2 - (7)^2$
 $(x+7)(x-7)$

(b) $x^2 - 4x - 21$

~~$x^2 + 3x + 7$~~

(c) $x^2 + 3x - 7x - 21$
 $x(x+3) - 7(x+3)$
 $(x-7)(x+3)$

∴ CF :- $(x-7)$

Q: 2(vii)

A: 2(viii)

$$\frac{2x+3}{5} = \frac{3-4x}{8}$$

$$5(3-4x) = 8(2x+3)$$

$$15 - 20x = 16x + 24$$

$$15 - 24x = 16x + 20x$$

$$-9x = 30x$$

$$-9x = x$$

$$36x$$

$$-\frac{1}{4} = x$$

Q: 2(in)

A: 2(in)

$$|4x-3| = 13$$

$$4x-3 = 13, 4x-3 = -13$$

$$4x = 13+3, 4x = -13+3$$

$$4x = 16, 4x = -10$$

$$x = \frac{16}{4}, x = \frac{-10}{4}$$

$$\left\{ 4, -\frac{5}{2} \right\}$$

Q. 2 (ii)

A. 2 (ii)

A(1, -2) B(1, 8) C(4, 3)

$$d_{AB} = \sqrt{(7-1)^2 + (8-(-2))^2}$$

$$= \sqrt{(7+1)^2 + (8+2)^2}$$

$$= \sqrt{(6)^2 + (10)^2}$$

$$= \sqrt{36 + 100}$$

$$= \sqrt{136}$$

$$= 2\sqrt{34}$$

$$d_{BC} = \sqrt{(4-1)^2 + (3-8)^2}$$

$$= \sqrt{(-3)^2 + (-5)^2}$$

$$= \sqrt{9 + 25}$$

$$= \sqrt{34}$$

$$d_{AC} = \sqrt{(4-1)^2 + (3-(-2))^2}$$

$$= \sqrt{(3)^2 + (3+2)^2}$$

$$= \sqrt{9 + (5)^2}$$

$$= \sqrt{9 + 25}$$

$$= \sqrt{34}$$

$$AB = BC + AC$$

$$2\sqrt{34} = \sqrt{34} + \sqrt{34}$$

$$2\sqrt{34} = 2\sqrt{34}$$

collinear.

$$Q: 2(ni)$$

$$A: 2(ni)$$

$$A(-1, 7) \quad B(7, -3)$$

midpoint P of AB:

$$\frac{-1+7}{2}, \frac{7+(-3)}{2}$$

$$\frac{6}{2}, \frac{7-3}{2}$$

$$\frac{6}{2}, \frac{4}{2}$$

$$3, 2$$

$$K = 2$$

$$P(3, 2)$$

$$d_m := \sqrt{(-1-3)^2 + (7-2)^2}$$

$$= \sqrt{(-4)^2 + (5)^2}$$

$$= \sqrt{16 + 25}$$

$$\sqrt{41} \text{ units.}$$